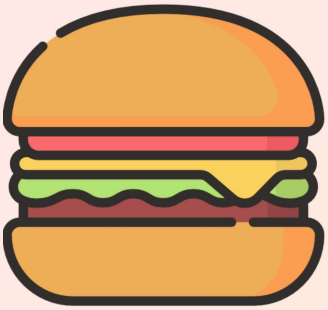
Candy Factory

The Number Systems

Class 5 Mathematics



CHAPTER - 1

CANDY - 1

The Number Systems

Indian System Chart

Crores		Lakhs		Thousands		Ones		
TC	C	TL	L	T-Th	Th	H	T	O
1	2	3	4	5	6	7	8	9

COMMA RULE: 3, 2, 2, 2 (START WITH 3 FROM RIGHT)

International System Chart

Millions			Thousands			Ones		
HM	TM	M	H-Th	T-Th	Th	H	T	O
1	2	3	4	5	6	7	8	9

COMMA RULE: ALWAYS 3, 3, 3 (GROUP IN 3S FROM RIGHT)

How to solve:

1. Place the number in the chart above starting from 'Ones' place.
2. Group the digits using commas according to the system rules.
3. Read the group names (Periods) like Millions, Thousands, etc.
4. Example: 123,456,789 is 'One hundred twenty-three million...'

EXERCISE 1: NUMBER NAMING FACTORY

1. Write Indian Name for: 75,42,10,500
2. Write International Name for: 999,999,999
3. Identify the Place of '5' in 45,12,00,000 (Indian)
4. Identify the Place of '2' in 123,456,789 (Int)
5. Write the number: Five Crore, Two Lakh, Fifty
6. Write the number: Twelve Million, Four Hundred
7. 78,00,00,005 in Indian System
8. 567,123,000 in International System
9. Name the Period for '8' in 8,45,00,000
10. Write in words: 101,101,101

CANDY - 2

Roman Numerals

ROMAN SYMBOLS

I	V	X	L
1	5	10	50

C	D	M
100	500	1000

THE RULES

Rule 1 (Add): If smaller is after bigger, add.
Ex: VI = 5 + 1 = 6.

Rule 2 (Sub): If smaller is before bigger, sub.
Ex: IV = 5 - 1 = 4.

Rule 3 (Max): Never repeat I, X, C more than 3 times!

Roman Table (1 - 50)

1	2	3	4	5	6	7	8	9	10
I	II	III	IV	V	VI	VII	VIII	IX	X
11	12	13	14	15	16	17	18	19	20
XI	XII	XIII	XIV	XV	XVI	XVII	XVIII	XIX	XX
21	22	23	24	25	26	27	28	29	30
XXI	XXII	XXIII	XXIV	XXV	XXVI	XXVII	XXVIII	XXIX	XXX
31	32	33	34	35	36	37	38	39	40
XXXI	XXXII	XXXIII	XXXIV	XXXV	XXXVI	XXXVII	XXXVIII	XXXIX	XL
41	42	43	44	45	46	47	48	49	50
XLI	XLII	XLIII	XLIV	XLV	XLVI	XLVII	XLVIII	XLIX	L

How to convert: 94

1. Break the number: 90 + 4
2. Convert 90: It is 10 less than 100 (100-10), so write XC.
3. Convert 4: It is 1 less than 5 (5-1), so write IV.
4. Combine: XCIV

EXERCISE 2: ROMAN ARENA

1. Write 42 in Roman
2. Write 99 in Roman
3. Write 55 in Roman
4. Write 14 in Roman
5. Write 88 in Roman
6. Write 73 in Roman
7. Write 29 in Roman
8. Write 48 in Roman
9. Write 64 in Roman
10. Write 100 in Roman
11. What is XXXVI?
12. What is XLIX?
13. What is LVIII?
14. What is XCII?
15. What is XIV?
16. Compare: XIX [ ] XXI
17. Compare: LX [ ] XL
18. Compare: XCI [ ] IC (Wait, is IC valid?)
19. Roman for 25
20. Number for LXXX

CANDY - 3

Stretching Machine

3 Stretching Examples

4,50,678

`4,00,000 + 50,000 + 0 + 600 + 70 + 8`

Vice Versa: 5,00,00,000 + 4,000 + 2

`5,00,04,002 (Watch out for missing zeros!)`

Steps to Expand:

1. Take the first digit from left.
2. Count how many digits are behind it. Put that many zeros.
3. Repeat for all digits, adding a '+' between them.
4. Skip digits that are '0' or write them as single '0'.

EXERCISE 3: ELASTIC NUMBERS

1. Expand 12,34,567
2. Expand 90,09,00,009
3. Expand 55,555
4. Expand 1,00,00,001
5. Expand 48,20,305
6. Short form: 4,00,000 + 2,000 + 5
7. Short form: 9L + 8T-Th + 1
8. Short form: 7C + 5
9. Short form: 10,000 + 1
10. Short form: 300 + 20 + 1

CANDY - 4

Identity Lab

The Zero Case

"In any number, at any place, the Place Value of 0 is ALWAYS 0 and Face Value of 0 is ALWAYS 0."

Master Conversion Table

Indian System	International	Roman-ish
1,00,000 (1 Lakh)	100,000 (100 Thousand)	---
10,00,000 (10 Lakh)	**1,000,000 (1 Million)**	---
1,00,00,000 (1 Crore)	10,000,000 (10 Million)	---
10,00,00,000 (10 Crore)	100,000,000 (100 Million)	---

EXERCISE 4: CONVERSION MASTER

1. Place value of 5 in 1,25,000
2. Face value of 9 in 9,87,654
3. 1 Million = [ ] Lakhs
4. 5 Crore = [ ] Million
5. How many zeros in 10 Million?
6. Position of 8 in 48,00,000
7. Place value of 0 in 506
8. Diff of P.V and F.V of 2 in 125
9. 100 Lakh = [ ] Crore
10. 20 Million = [ ] Crore

THINK AND TRY!

Hard Candy Challenges

LOGIC

Write a 9-digit number that remains the same even if written in reverse.

ROMAN

Is 'IL' a valid Roman Numeral for 49? Explain why (Hint: Rule 2 limits).

MONEY

If you have 1 Billion candies, how many Crores is that? (Hint: 1 Billion = 1000 Million)

PLACE

Find a number where the Place value of 4 is 4,000 and Face value is 4.

Quick Tips & Tricks

- 10 Lakhs = 1 Million (Just move the commas!)
- Place value = Digit times of Place.
- V, L, D are never subtracted or repeated!
- Comma trick: International is always "Three-Three-Three".

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